

From Conventional Data Warehousing to Lakehouse Architecture: A Comparative Analysis of Databricks Lakehouse and Data Warehouse Systems for Scalable Analytics

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Abstract—The rapid growth of enterprise data has increased the demand for data architectures capable of supporting diverse data types and analytical workloads. Data warehouse systems have long been used to support structured data analytics and business intelligence applications. However, modern organizations increasingly require platforms that can manage structured, semi-structured, and unstructured data while supporting advanced analytics and machine learning workloads. This paper examines data warehouse systems and the Databricks Lakehouse architecture by analyzing their architectural design, key characteristics, and performance features. A comparative analysis is conducted based on scalability, flexibility, processing capabilities, cost efficiency, and analytical support. The discussion highlights the different approaches adopted by both architectures in addressing modern data management requirements. The findings provide insights into the strengths, limitations, and trade-offs associated with each architecture, allowing readers to evaluate their suitability according to specific business and technical requirements.

Index Terms—Databricks Lakehouse, Data Warehouse, Big Data Architecture, Delta Lake, Scalable Analytics

I. INTRODUCTION

The rapid growth of data generated by modern enterprises has significantly transformed the requirements of data processing and analytics systems (Gandomi & Haider, 2015). Organizations increasingly require platforms capable of handling structured, semi-structured, and unstructured data while supporting both batch and real-time analytical workloads. Conventional data warehouse systems were designed to address structured data processing with high reliability and performance (Inmon, 2005). However, these systems face limitations in managing the volume, velocity, and variety of modern data (Gandomi & Haider, 2015).

A major challenge in conventional architectures is the separation between data storage and advanced analytics systems, which leads to data duplication, increased operational complexity, and delays in data processing workflows (Armbrust et al., 2021). These limitations have driven the development of modern data architectures, such as the Lakehouse, particularly as implemented on platforms like Databricks. Lakehouse architecture aims to unify the capabilities of data lakes and

data warehouses into a single system, enabling scalable and flexible data management (Armbrust et al., 2021).

This paper aims to analyse and compare conventional data warehouse systems and the Databricks Lakehouse architecture. The study examines their architectural design, key characteristics, and real-world applications. A comparative analysis is then conducted to evaluate how each architecture addresses modern data processing and analytics requirements.

II. ARCHITECTURE AND ANALYSIS

A. Conventional Data Warehouse Overview

Conventional data warehouses are designed to store structured data for reporting and analysis. Data is collected from multiple operational systems and processed through an Extract, Transform, Load (ETL) pipeline, where it is cleaned and transformed before being stored in a predefined format (Inmon, 2005). This ensures data consistency and supports efficient querying for business intelligence applications.

The architecture typically consists of data sources, a staging area, ETL pipelines, centralized storage, and query processing systems. As illustrated in Figure 1, data flows from source systems into a staging area where it is temporarily stored and processed before being loaded into a structured data warehouse. This design follows a schema-on-write approach, which improves query performance but limits flexibility when handling diverse data types (Kimball & Ross, 2013). The architecture also includes data marts, which provide specialized data for different business functions such as sales and inventory analysis (Kimball & Ross, 2013).



Figure 1: Conventional Data Warehouse Architecture
(adapted from Estuary, 2024, based on Inmon, 2005)

Furthermore, the reliance on batch processing introduces latency, and scaling infrastructure often results in high operational costs (Armbrust et al., 2021). Despite these limitations, conventional data warehouses remain effective for structured analytics in environments with stable data requirements.

B. Databricks Lakehouse Architecture

The Databricks Lakehouse architecture is a modern data management approach that combines the scalability of data lakes with the reliability and performance of data warehouses. It enables organizations to store, process, and analyze structured, semi-structured, and unstructured data within a unified platform (Armbrust et al., 2021). Unlike conventional architectures that separate storage and analytics systems, Lakehouse integrates these components, allowing multiple analytical workloads, including business intelligence and machine learning, to operate on the same data.

The architecture is structured into multiple layers, including ingestion, storage, metadata, processing, and consumption components. Data is first collected from various sources through batch or streaming pipelines and stored in scalable cloud storage using open formats such as Parquet (Armbrust et al., 2020). A metadata layer, typically implemented using technologies such as Delta Lake, ensures data consistency by providing schema enforcement and ACID transaction support (Armbrust et al., 2020). These capabilities improve data reliability while maintaining the flexibility of a cloud-based storage system. Processing engines such as Apache Spark enable data transformation, querying, and analytics (Zaharia et al., 2016), while APIs allow interaction between the data platform and analytical tools. Finally, the consumption layer delivers insights through dashboards, reports, and machine learning models, supporting large-scale data-driven applications (Abadi, 2009).

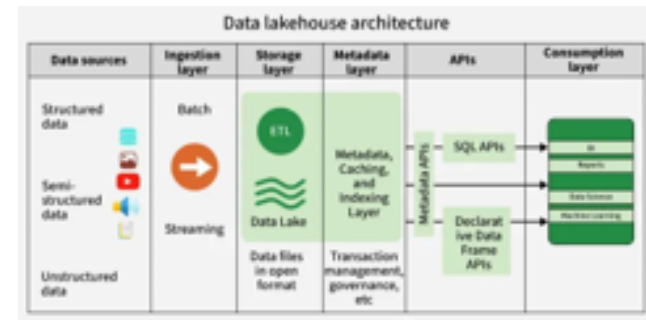


Figure 2: Databricks Lakehouse Architecture

(adapted from GeeksforGeeks, 2024, based on Armbrust et al., 2021)

As shown in Figure 2, Lakehouse architecture integrates data ingestion, storage, and analytics into a unified system, enabling scalable and flexible data processing across multiple workloads. The differences between these architectures become more apparent when analyzed across key performance and design criteria.

C. Comparative Analysis: Data Warehouse vs Databricks Lakehouse

The transition from conventional data warehouse systems to Lakehouse architecture reflects changing requirements in modern data environments. Conventional data warehouses and Lakehouse platforms adopt different approaches to data storage, processing, and analytics. Data warehouses are primarily designed for structured analytical workloads, whereas Lakehouse architecture supports a broader range of data types and processing requirements. These architectural differences influence factors such as scalability, flexibility, processing capabilities, and system complexity. Therefore, a comparison across key evaluation criteria is necessary to understand the characteristics and trade-offs associated with each approach.

In contrast, the Databricks Lakehouse architecture enables unified analytics by integrating storage, processing, and analytical capabilities within a single platform. This allows organizations to handle large-scale and diverse datasets more efficiently. However, the increased flexibility introduces challenges related to system complexity and management.

TABLE I
COMPARATIVE ANALYSIS OF DATA WAREHOUSE AND DATABRICKS LAKEHOUSE ARCHITECTURES

Criteria	Data Warehouse	Databricks Lakehouse
Data Type Support	Primarily structured data	Supports structured, semi-structured, and unstructured data
Schema Approach	Schema-on-write	Schema-on-read with enforcement
Processing	Batch processing	Batch and real-time streaming
Scalability	Limited scalability	Highly scalable (cloud-based)
Machine Learning	Limited support	Integrated ML capabilities
Cost Efficiency	High infrastructure cost	Lower cost with cloud storage
Complexity	Lower complexity	Higher system complexity

The comparison indicates that both architectures adopt different approaches to data storage, processing, and analytics. Data warehouse systems emphasize structured data management and predefined schemas, whereas Lakehouse architecture supports a wider variety of data types and processing methods. These differences influence scalability, flexibility, governance requirements, and implementation complexity. For example, the schema-on-write approach commonly used in data warehouse systems requires data structures to be defined before data is loaded, while Lakehouse architecture supports greater flexibility through its support for multiple data formats and processing frameworks. Consequently, the suitability of each architecture depends on organizational objectives, workload characteristics, and data management requirements (Armbrust et al., 2021).

D. Real-World Application

A real-world example of large-scale data processing systems can be observed in Netflix, a global streaming platform that processes massive volumes of user interaction data. Netflix utilises distributed data processing frameworks such as Apache Spark to support large-scale analytics and personalised content recommendation systems (Zaharia et al., 2016).

Data is continuously ingested from multiple sources, including user activity logs, streaming data, and operational systems (Zaharia et al., 2016). The platform supports both batch and real-time data processing, enabling timely analysis of user behaviour and system performance. This approach reflects key principles of modern data architectures, where scalable storage and distributed processing are integrated to support efficient data workflows. These characteristics share similarities with principles commonly associated with modern Lakehouse architectures, including scalable storage, distributed processing, and support for advanced analytics (Armbrust et al., 2021). In addition, the integration of analytics and machine learning capabilities allows Netflix to generate insights directly from large datasets, supporting applications such as recommendation systems and demand forecasting.

However, implementing such large-scale systems also introduces challenges. While distributed data platforms improve scalability and performance, they require careful resource management and robust data governance to ensure system reliability (Abadi, 2009). This case study demonstrates how real-world data platforms adopt architectural principles aligned with the Lakehouse model while also presenting practical trade-offs.

This demonstrates how modern data platforms are evolving towards unified architectures like the Lakehouse model, where data storage, processing, and analytics are integrated to support scalable and efficient data workflows.

III. CONCLUSION

This paper has examined the evolution of data processing systems from conventional data warehouse architectures to the Databricks Lakehouse model. Conventional data warehouses provide a structured and reliable environment for data storage and querying, particularly for business intelligence applications. However, their limitations in handling diverse data types and supporting real-time analytics highlight challenges in modern data environments.

The Databricks Lakehouse architecture introduces a unified approach that integrates data storage, processing, and analytics within a single platform. This enables the management of structured, semi-structured, and unstructured data while supporting both batch and real-time workloads. The comparative analysis suggests that conventional data warehouses provide a stable environment for structured analytics, whereas Lakehouse architecture offers greater flexibility in supporting diverse data workloads.

Despite these advantages, the Lakehouse architecture presents limitations related to governance, resource management, and implementation complexity. These challenges indi-

cate that the adoption of modern data architectures requires careful consideration of system requirements and organizational capabilities.

However, the Lakehouse paradigm remains an evolving architecture, and its long-term performance and governance effectiveness in highly regulated enterprise environments require further empirical validation.

Future developments in data architecture may focus on improving integration between data systems, enhancing data governance frameworks, and optimizing performance for large-scale distributed environments. Such advancements are expected to further refine the balance between flexibility, scalability, and system manageability in modern data processing systems.

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